

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

- Motion in plane ✓
- Newton laws ✓
of motion
- Baclogs

3lec ✓

WORK, ENERGY, AND POWER

LECTURE-1

PHYSICS

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Topics To Be Covered

- 1 Work ✓
- 2 Work done by variable force ✓
- 3 Energy ✓
- 4 Kinetic Energy ✓
- 5 Work Energy Theorem ✓ (Numericals) ✓

“INDEED, WITH
HARDSHIP COMES
EASE.”

WORK, ENERGY & POWER

Work done by a constant force and a variable. force;
Kinetic energy, work energy theorem, Power. Notion
of Potential Energy, potential energy of a spring,
conservative forces: non-conservative forces, motion
in vertical circle; Elastic and inelastic collisions in
one and two dimensions.

WORK Energy and power

⇒ 9th

□ → work

→ Types of work

→ Energy

→ Types of Energy

→ $\underline{K.E} = \frac{1}{2}mv^2$

→ $PE = mgh$

} ME

→ Work Energy theorem

• PE Energy in spring

• Vertical Circular Motion

• Conservation and Non-Conservative force

• Collision

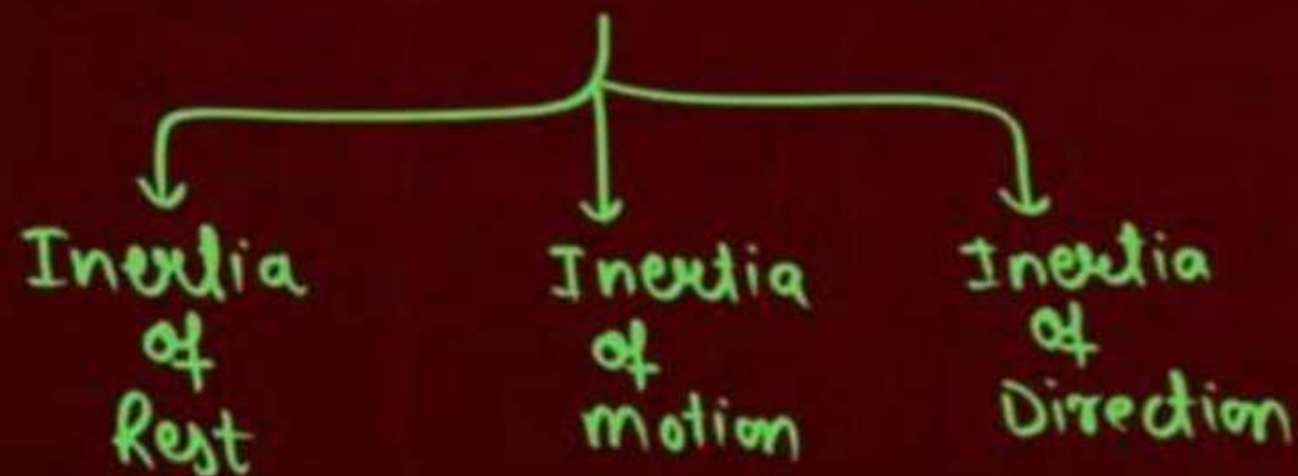
Elastic

Inelastic

Recap From 1st Lecture

Inertia (1st law of motion)

Property of Body by virtue of which Body cannot change its state of **Rest**, **motion** and **Direction**



Momentum (p)

$$\vec{p} = m\vec{v} \quad \text{kg m/s Vector}$$

$$\bullet p_f = mv \quad \bullet p_i = mv$$

② # Newton 2nd laws of Motion

$$F = \frac{\Delta p}{\Delta t} = ma \rightarrow \text{qth}$$

$$F = \frac{\Delta p}{\Delta t} = \frac{dp}{dt} = m \frac{dv}{dt} + v \frac{dm}{dt}$$

③ **Impulse**

Force acting for the short period of time.

$$\uparrow F = \frac{\Delta p}{\Delta t} \downarrow$$

$$F = \frac{dp}{dt} \quad \{ F dt \} = dp$$

Impulse

Impulse momentum theorem

Recap From 2nd Lecture

Newton third law of motion

Action = -Reaction

- * Rocket propulsion
- * walking
- * Boat
- * Recoiling on gun
- * Flying of Bird.

law of conservation of momentum

if $F_{ext} = 0$

$$P_i = P_f$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Application of law of conservation of momentum

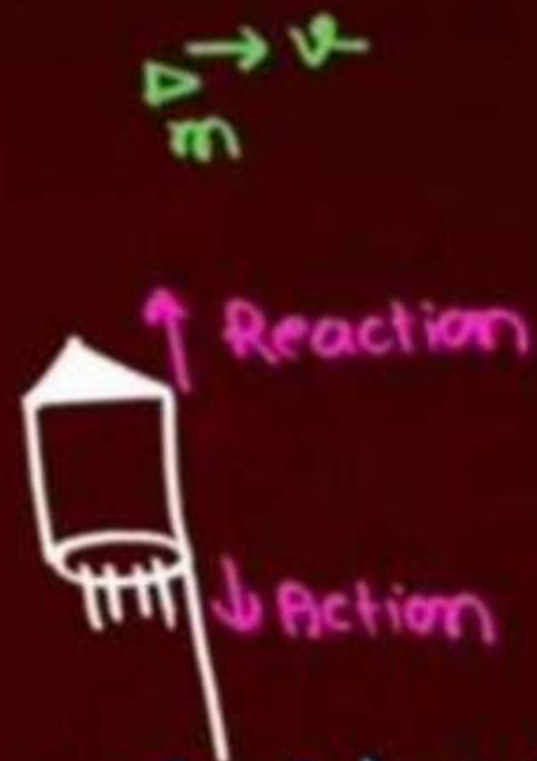
① Recoiling velocity of gun

$$V = -\frac{mv}{M}$$



② Rocket propulsion

$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$



? why Newton 2nd law is Maxwell's law of motion

$$F = \frac{dP}{dt}$$

2nd law derived $V = u$

3rd law derived $f_{12} = -f_{21}$

Lecture no. 3

Types of forces

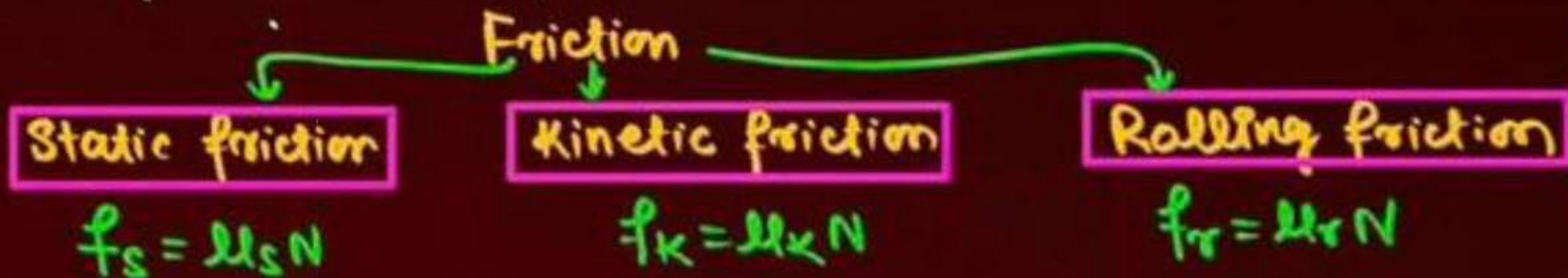
- * Gravitation force
- * Spring force
- * Tension
- * Normal Force
- * Electrostatic force
- * Friction force ✓

Friction

- * Force oppose the Relative Motion $f = \mu N$
- * Roughness & Adhesion

Laws of friction

- $f = \mu N$
- friction is independent of Area.

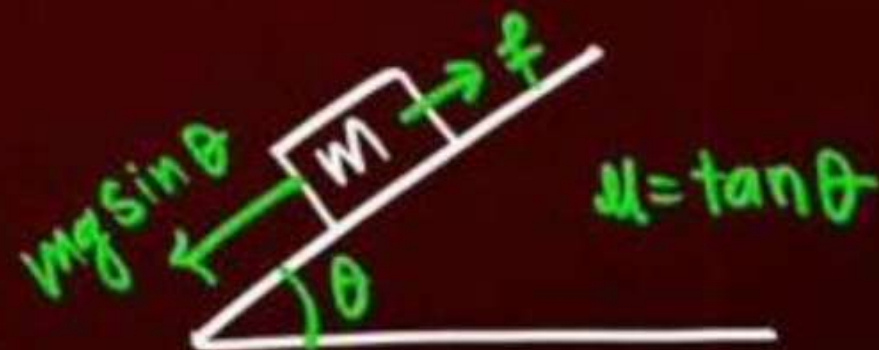


- ✓ Advantage of friction ✓
- ✓ Disadvantage of friction ✓
- ✓ How to Reduce the friction. ✓

Angle of friction

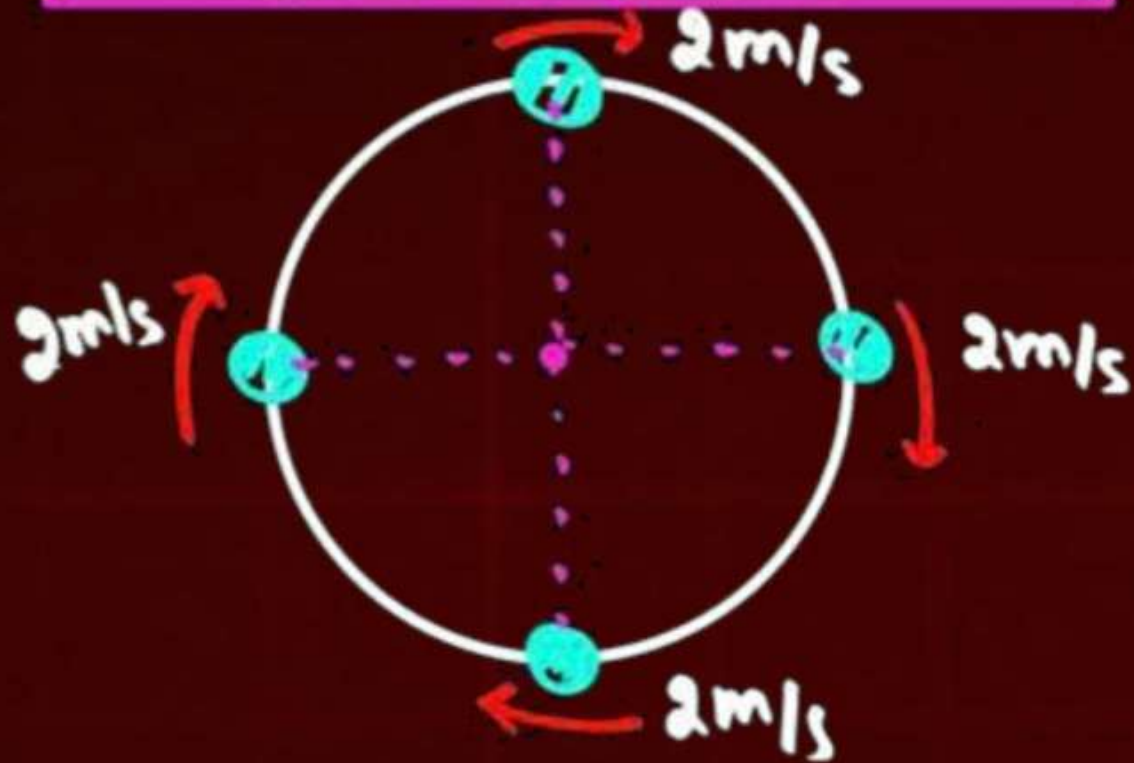


Angle of Repose



(Lecture no. 4)

• Centripetal acc/r



Speed constant

velocity change

$a_c = \frac{v^2}{r}$ $F_c = ma_c$
 $= m \frac{v^2}{r}$

① Turning on friction Road



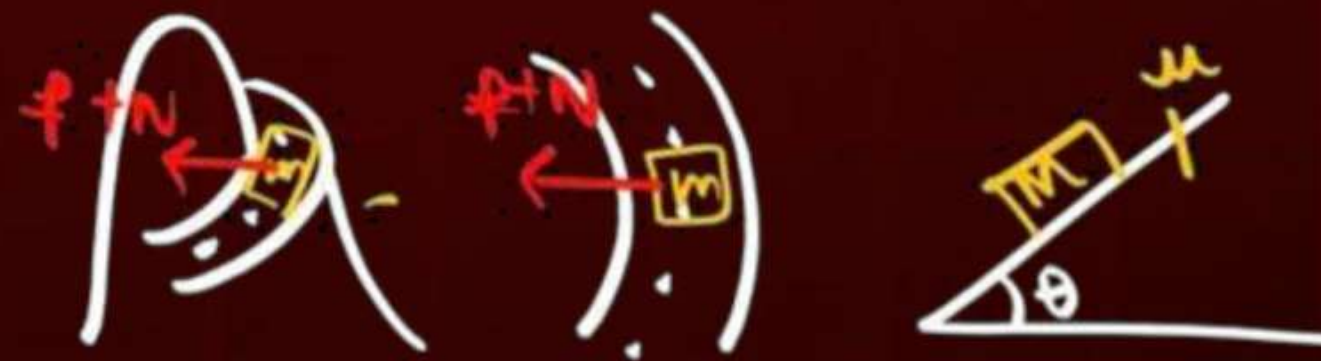
$$v = \sqrt{rg\mu}$$

② Turning on Banked Road



$$v = \sqrt{rg \tan \theta}$$

③ Turning on friction Road and Banked Road



$$v = \sqrt{rg \left(\frac{\mu + \tan \theta}{1 - \mu \tan \theta} \right)}$$

WORK

$$1\text{J} = 1000\text{erg}$$

Note Physics main jo work hota hai, wo Society ke work se Bilkul alag hota hai

CE → ME

Work is mode of transfer of Energy

Work done is the dot product of Force and displacement

$$W = \vec{F} \cdot \vec{x}$$

$$W = (Fx \cos \theta)$$

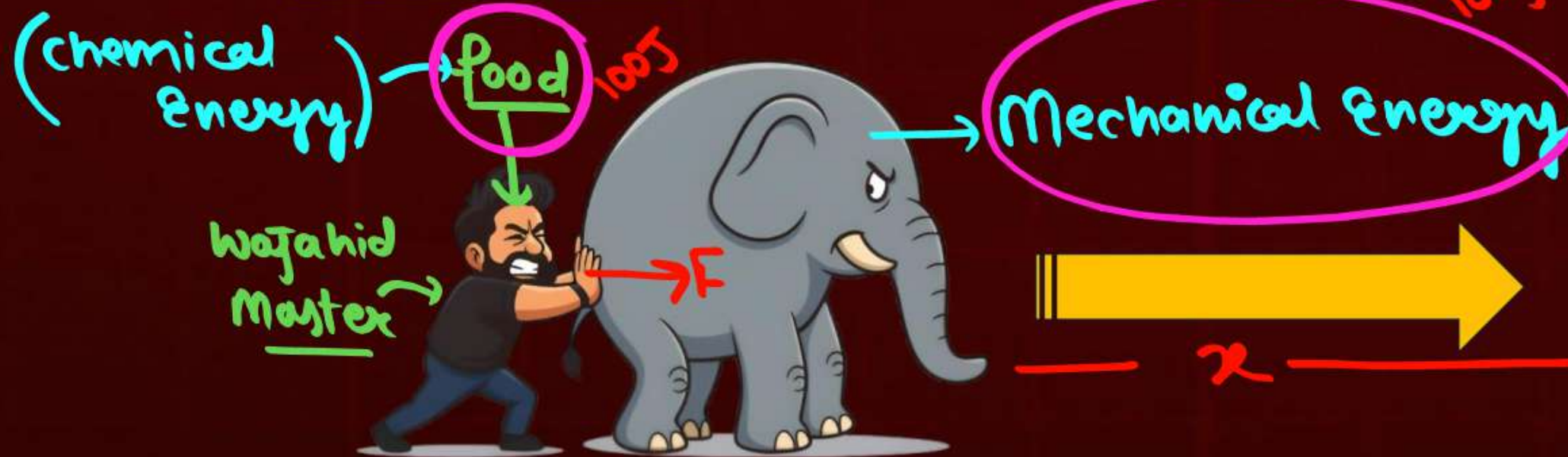
$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

Work is Scalar.

S.I unit → Joule

C.G.S unit → erg

$$1\text{J} = 10^7\text{erg}$$



Types of work

$$W = Fx \cos \theta$$

on the Basis of angle b/w $\vec{F}$ and $\vec{x}$

Positive work

Negative work

Zero

Types of work

on the Basis of Force

$$F = c$$

$$F = v$$

Work done By constant force $F = c$

Work done By variable force $F = v$

① POSITIVE WORK

- The positive work is said to be done when force and displacement are in same direction



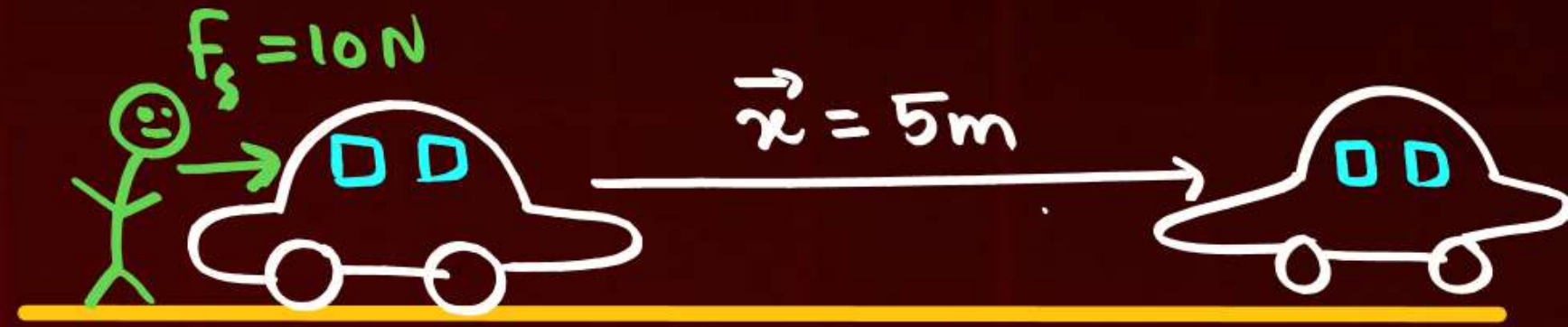
$$W = \vec{F} \cdot \vec{x}$$

$$W_s = F_s x \cos \theta$$

$$W_s = 10 \times 5 \cos 0^\circ$$

$$W_s = 50 \times 1 = +50 \text{ J} \checkmark$$

#



② NEGATIVE WORK

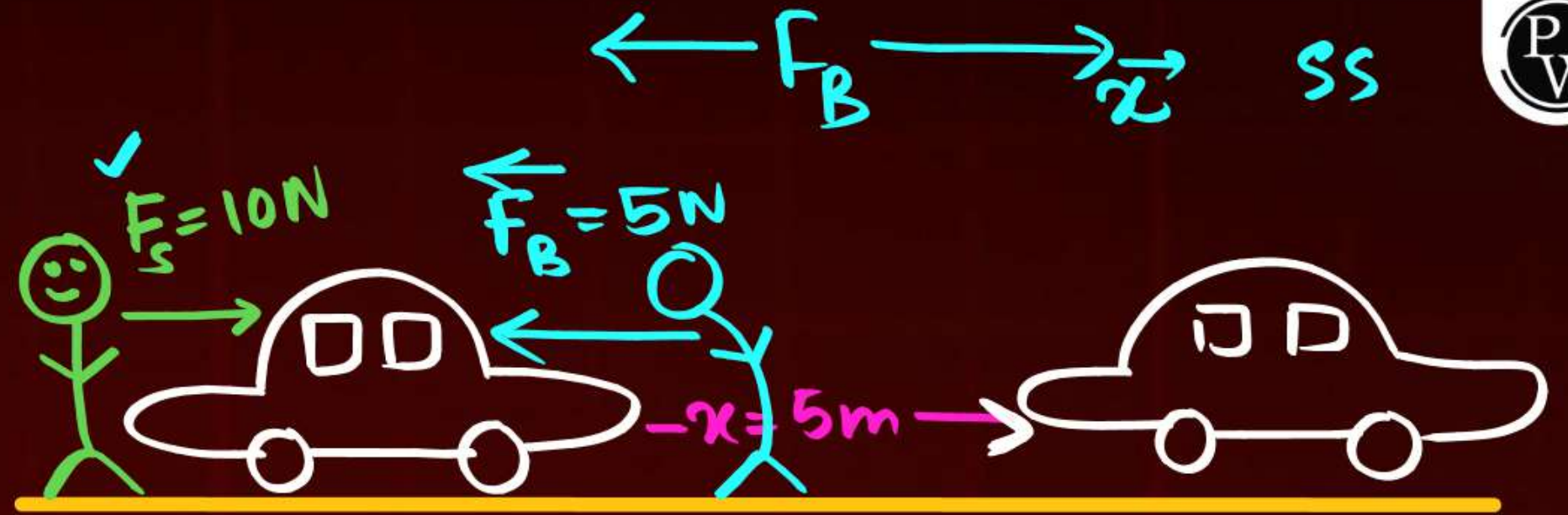
* when force $\vec{F}$ and displacement $\vec{x}$ are in opposite direction

$$W = \vec{F} \cdot \vec{x}$$

$$W_B = F_B x \cos \theta$$

$$W_B = 5 \times 5 (\cos 180^\circ)$$

$$W_B = 25 \times -1 = -25 \text{ J}$$

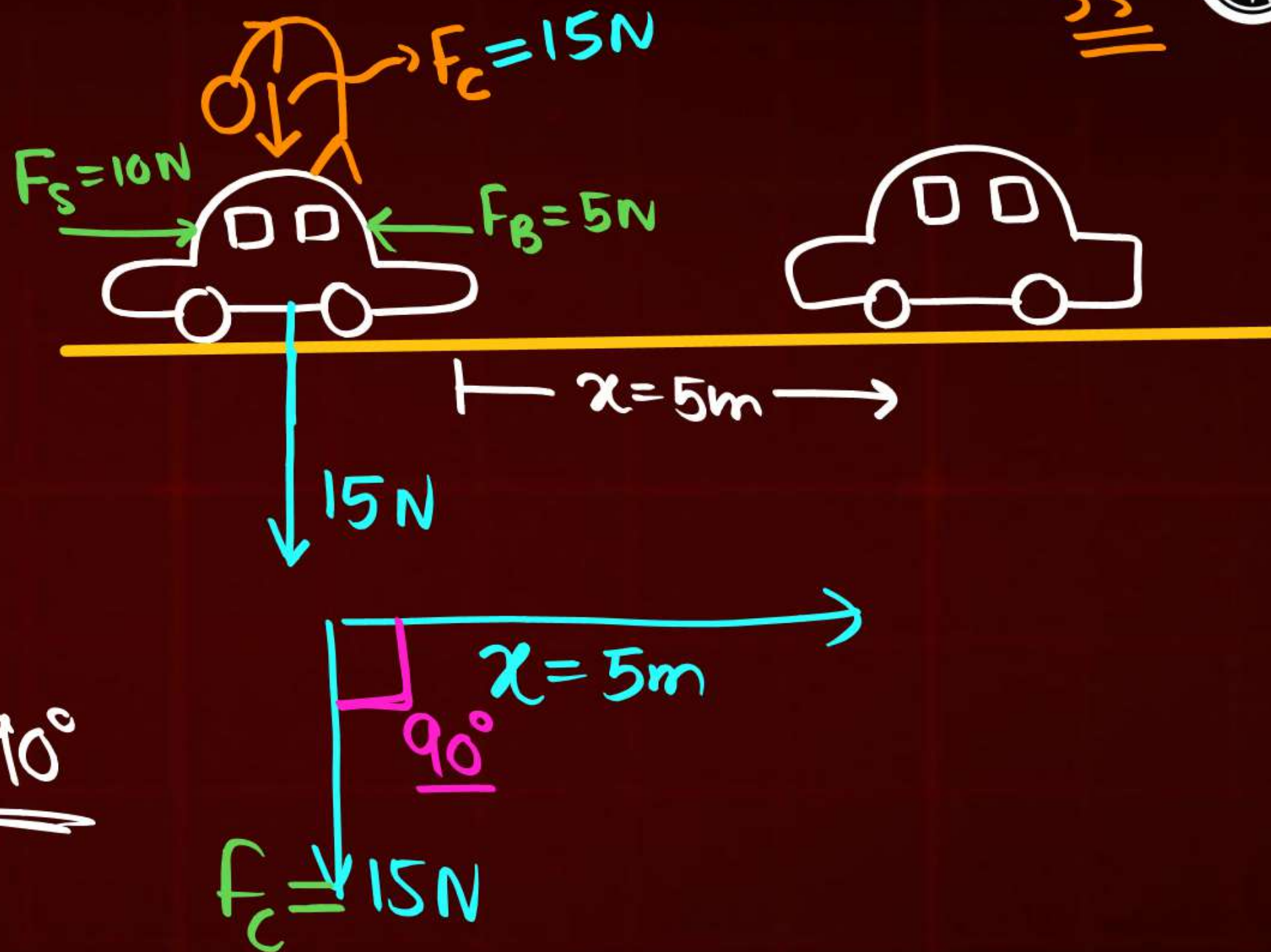


3) ZERO WORK

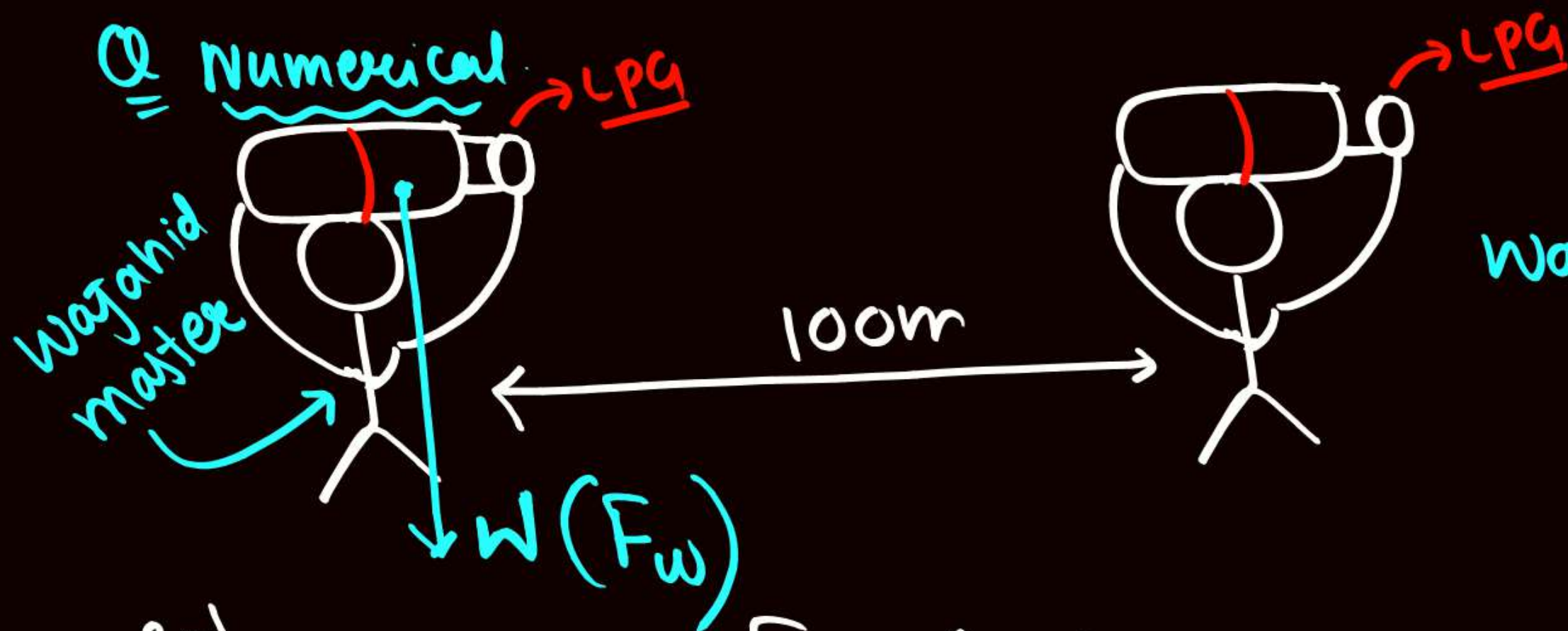
When force and displacement are perpendicular to each other

$$W = \vec{F} \cdot \vec{x}$$

$$\begin{aligned} W_c &= F_c x \cos \theta \\ &= 15 \times 5 \times \cos 90^\circ \\ &= 15 \times 5 \times \underline{0} \\ &= 15 \times 0 \\ &= 0 \text{ J} \end{aligned}$$



SS

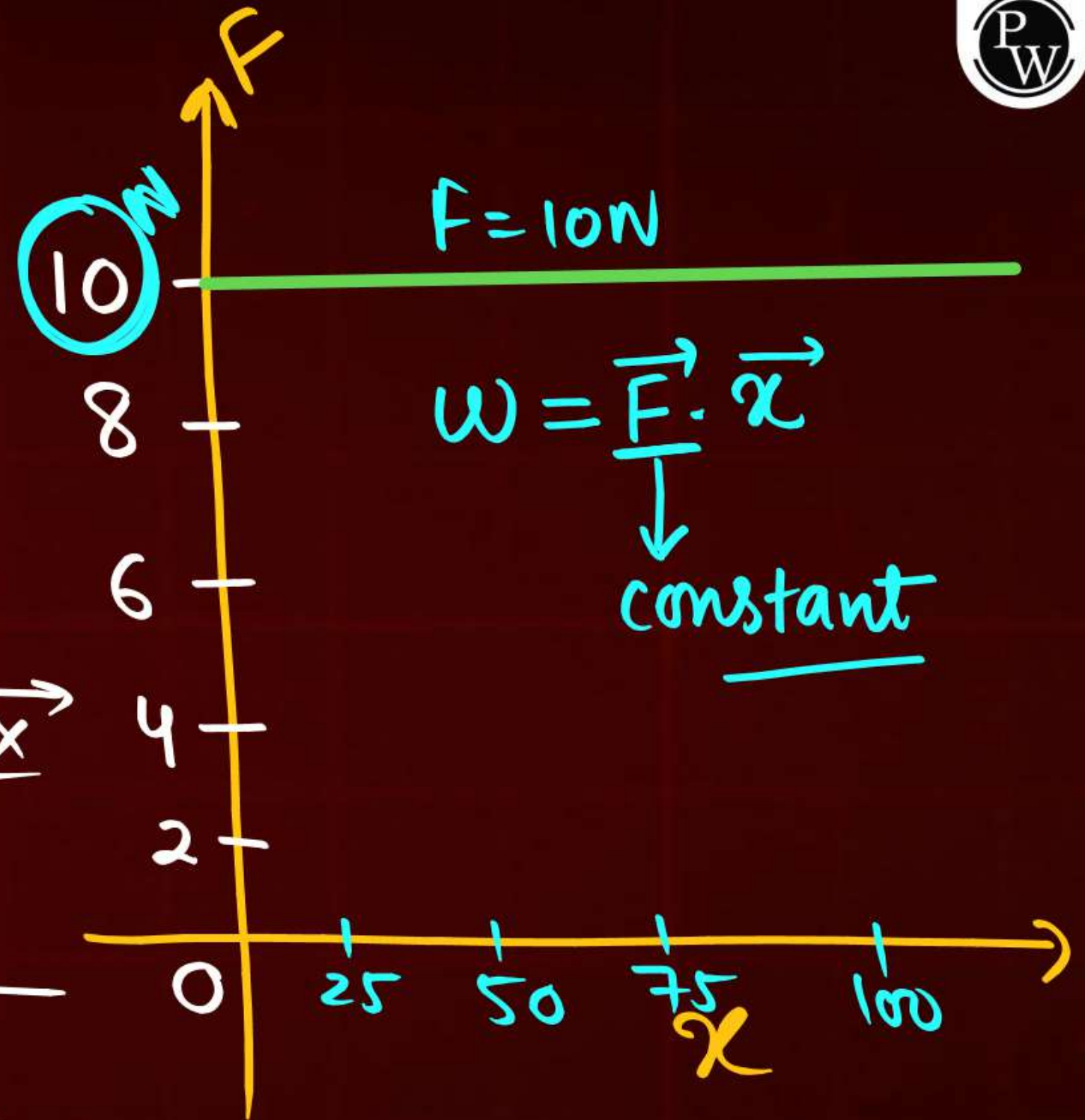
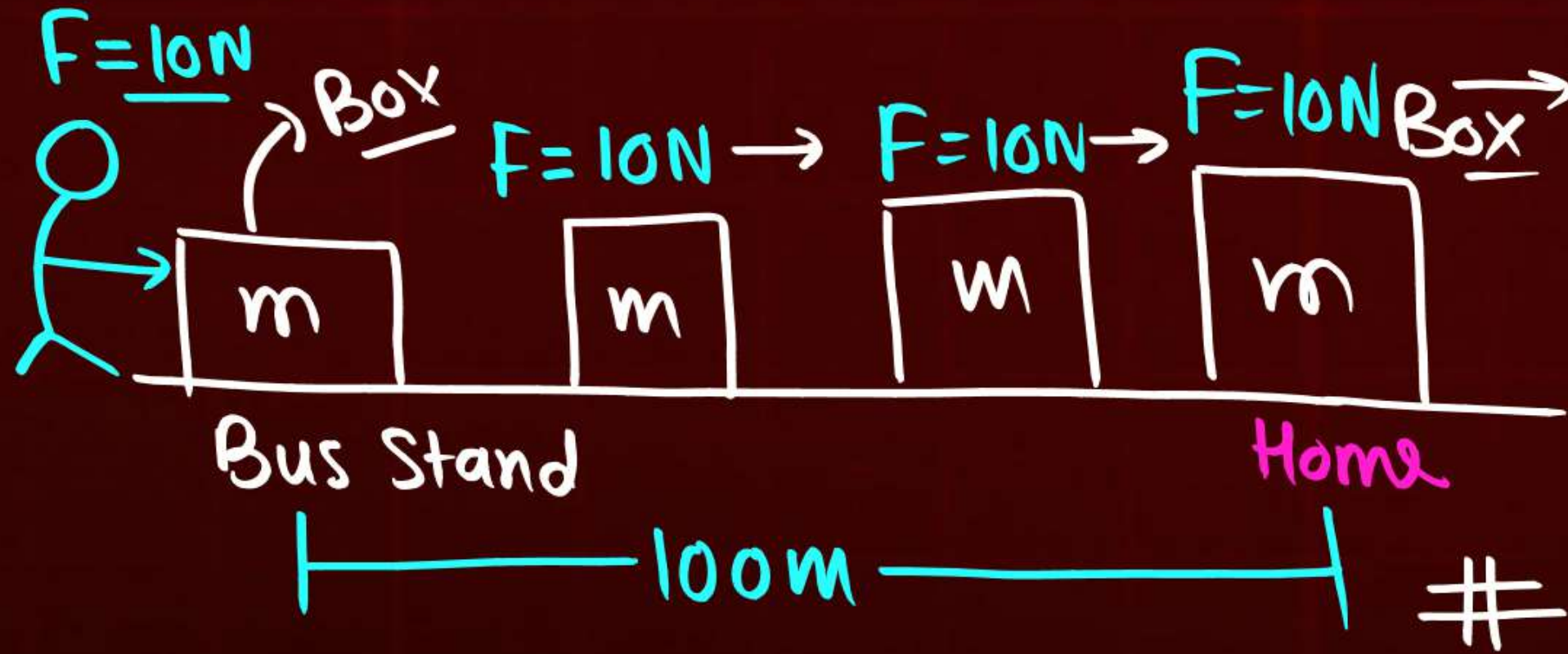


Sol

$$\begin{aligned} W &= F \cos \theta \\ &= W \cos 90^\circ \\ &= W \times 100 \times 0 \\ &= W \times 0 \\ &= 0 \text{ J} \end{aligned}$$

① Work done by Constant Force

The work in which **Force** is **Constant** is known as work done By constant Force



Sol

→ 10 N

→ 100 m

$$W = Fx \cos \theta^\circ$$

$$= 10 \times 100 \cos 0^\circ$$

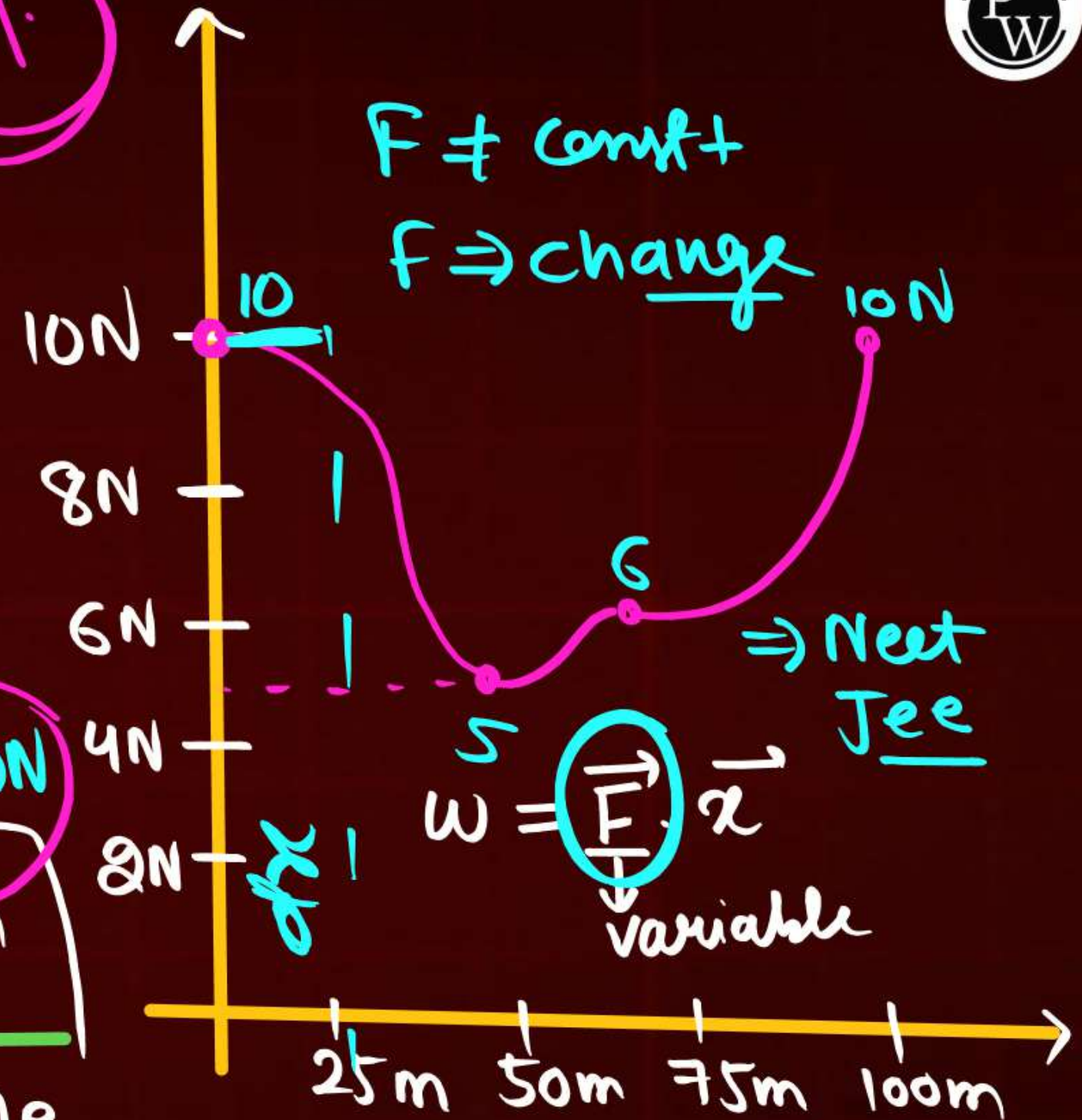
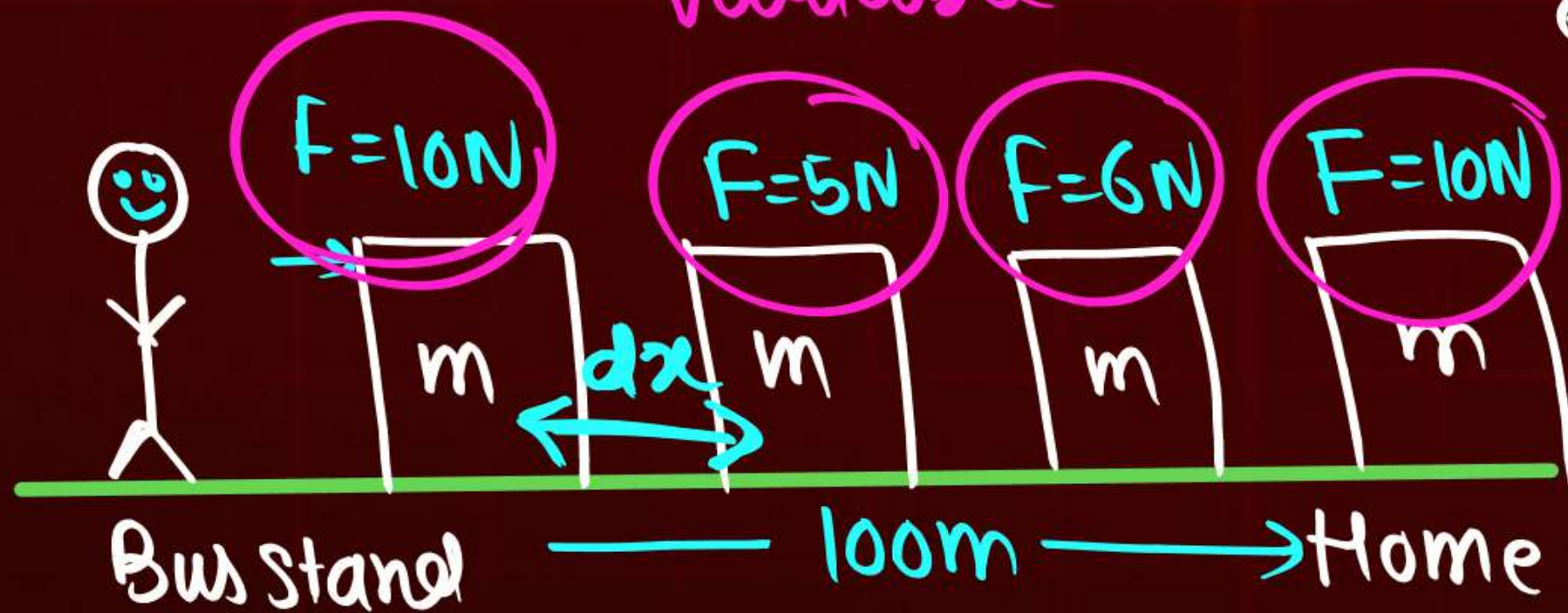
$$= 1000 \times 1$$

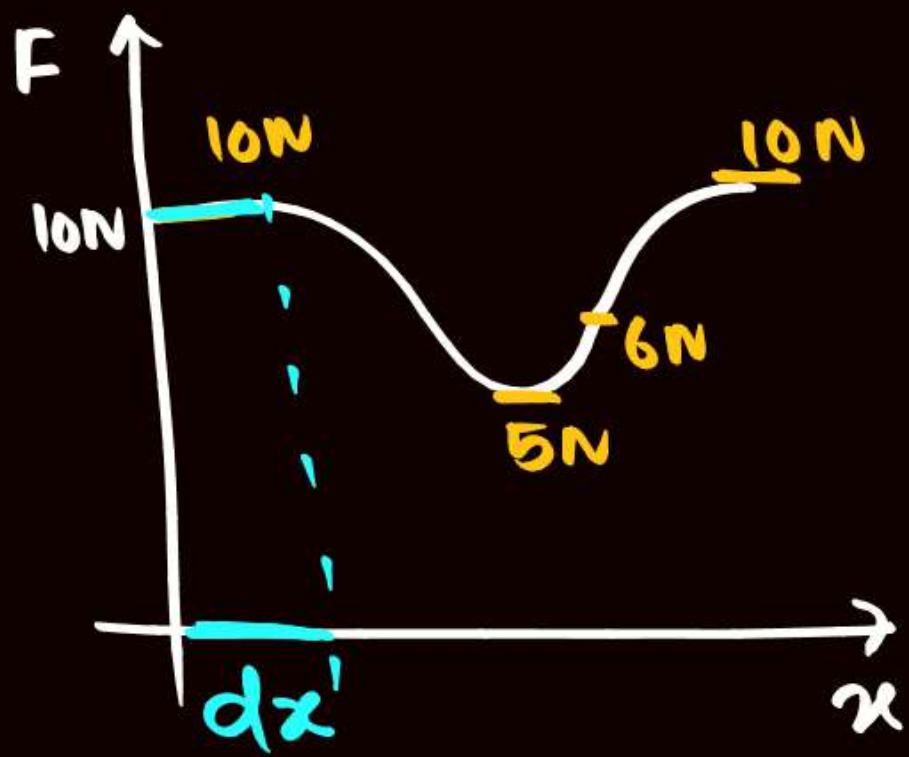
$$= \underline{1000 \text{ J}}$$

② Work done by Variable Force

when the force is changing
(variable) during work done

$F \rightarrow$ chang
variable





$$\# W = \vec{F} \cdot \vec{x} \times$$

10
5
5
10

$$dW = \vec{F} \cdot d\vec{x}$$

$$\int dW = \int \vec{F} \cdot d\vec{x}$$

Feel

$$\Rightarrow W = \int \vec{F} \cdot d\vec{x}$$

Note:- When the force is Variable, we calculate the work from integrated formula of $\vec{F} \cdot d\vec{x}$

$$\int dW = \int \vec{F} \cdot d\vec{x}$$

ENERGY

- The capacity of do work

Energy → Work

- Energy unit → Joule

- Scalar

S.I Unit

Types of Energy

TYPES OF ENERGY

01

Solar ✓

02

Wind
Energy ✓

03

Hydropower ✓

04

Petroleum ✓

05

Coal
Energy ✓

06

Tidal
Energy ✓

SOLAR

The energy you can get from sunlight. ✓



WIND ENERGY

The Energy you can get from Wind/Air



HYDROPOWER

The energy get from flowing water



PETROLEUM

The energy get from Petrol



COAL ENERGY

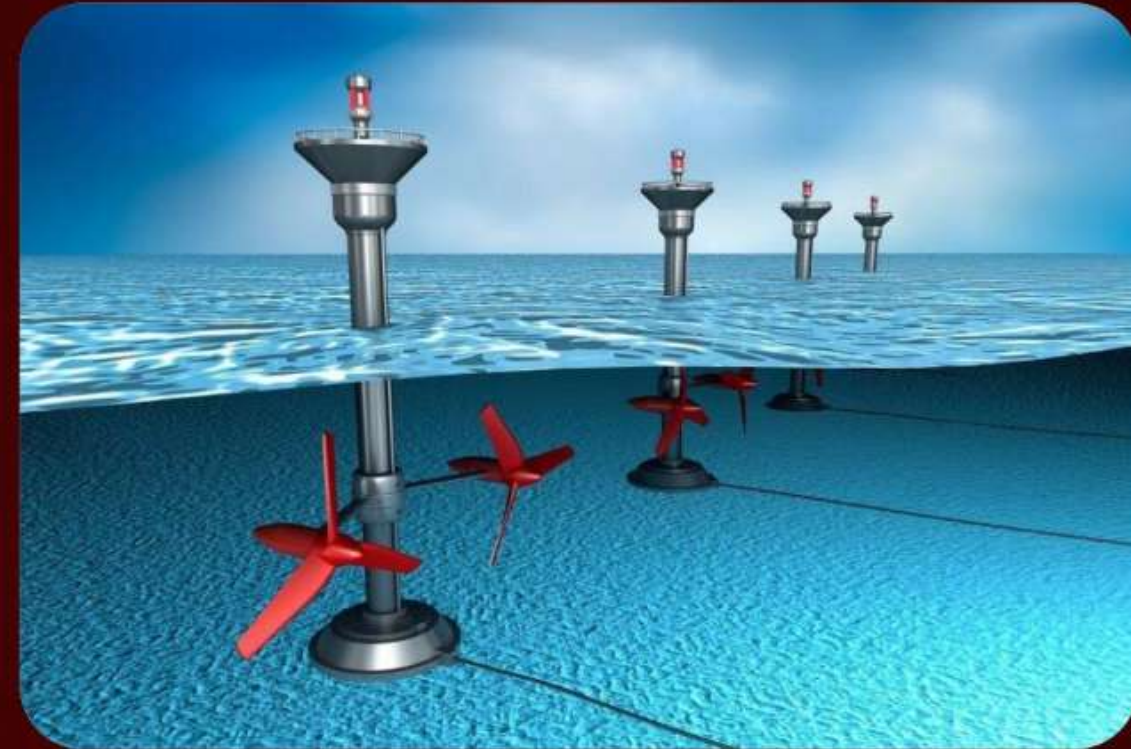
The Energy you can get from Coal ✓

Coal Samjhe koila 😊



TIDAL ENERGY ✓

The Energy you can get from the waves of the Oceans





$$ME = K.E + P.E$$

$$T.E = K.E + P.E$$

KINETIC ENERGY

(9th) ✓

- The Energy due to the velocity of the particle

Or

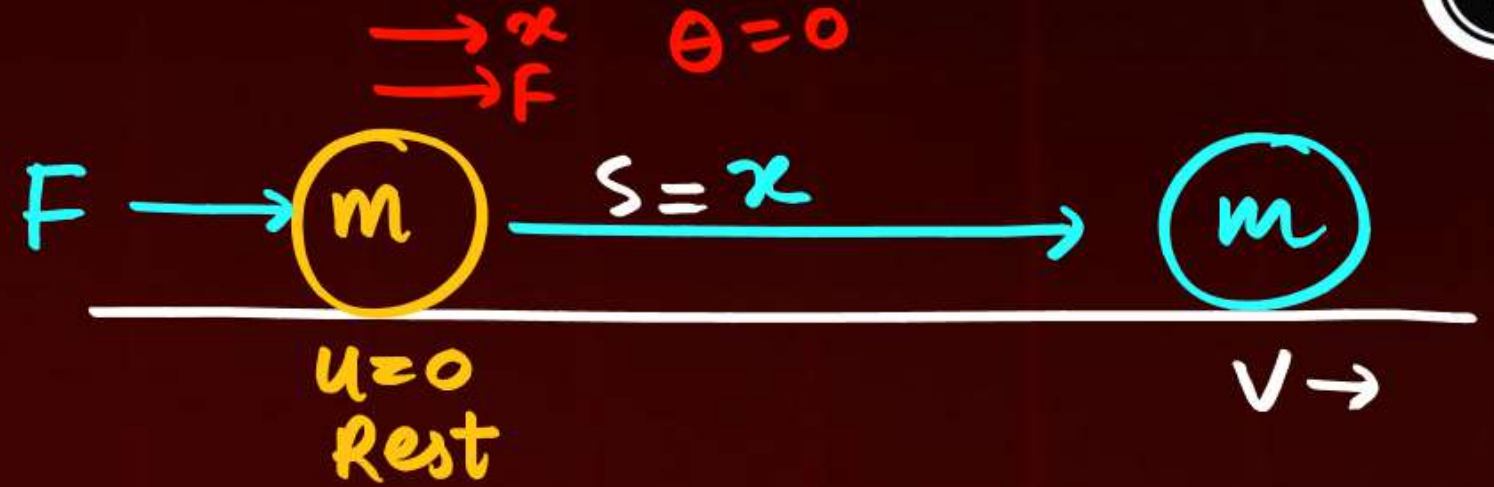
- The energy due to the motion of the particle

$$K \text{ or } K.E = \frac{1}{2}mv^2$$

$m \rightarrow$ mass

$v \rightarrow$ velocity (v)

Derivation



#

$$W = Fx \cos \theta$$

$$W = Fx \cos 0^\circ$$

$$W = Fx$$

$$W = max$$

$$W = m \cancel{a} \frac{v^2}{2\cancel{a}}$$

$$W = \frac{mv^2}{2}$$

Behn

Bhai

$$W = \frac{1}{2}mv^2$$

$$K.E = \frac{1}{2}mv^2$$

Newton 3rd Equation of Motion

$$v^2 = u^2 + 2as$$

$$v^2 = 0^2 + 2ax$$

$$v^2 = 2ax$$

$$\frac{v^2}{2a} = x$$

Neet

Another formula of acceleration

$$a = \frac{dv}{dt}$$

①

m/d $\rightarrow$ dx

$$a = \frac{dv}{dt} \times \frac{dx}{dx}$$

$$a = \frac{dv}{dx} \times \frac{dx}{dt}$$

$$a = \frac{dv}{dx} v$$

$$a = v \frac{dv}{dx}$$

②

Work Energy Theorm

Neet

✓ WORK Energy theorem

2025/24/23/22/21/19/18

cc Work is change in K.E

$$W = \Delta K.E$$

✓
Proof

Aaiya dekhte h kisme kitna hai Energy 😄 😄



SUMMARY

Work

Types

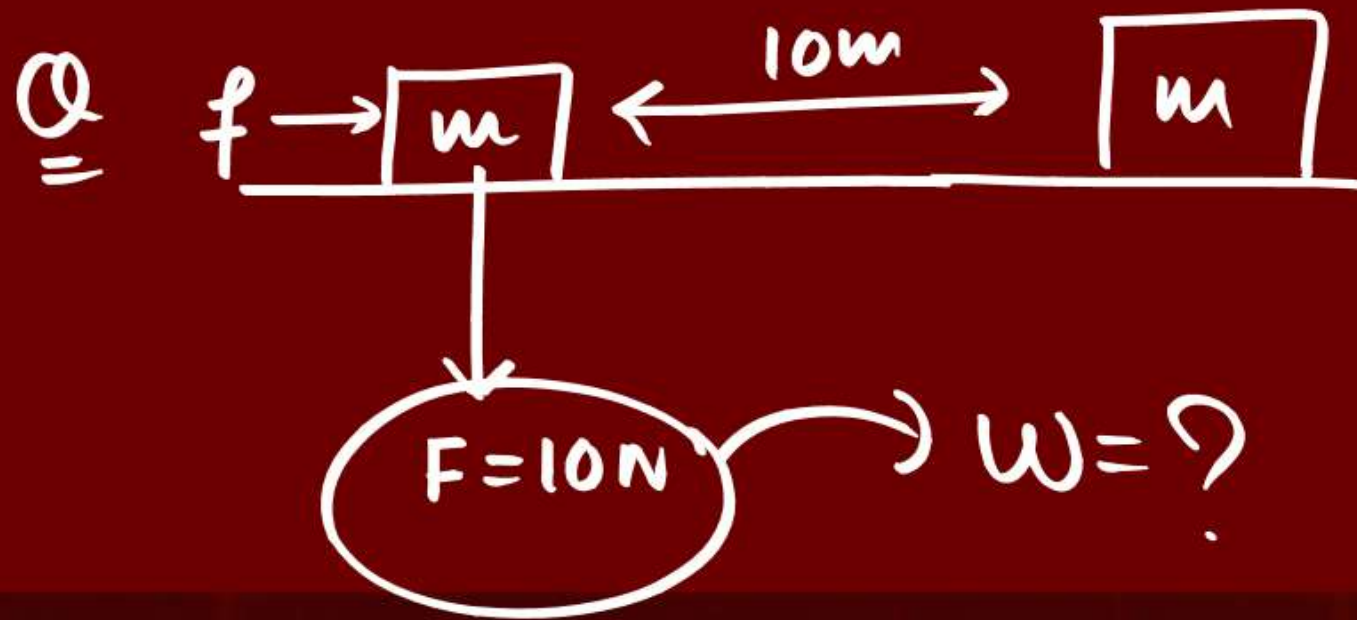
Energy

KE

$$W = \Delta K \checkmark$$



HOMework



Thank
You

